

DC – AI – Routing – Switching – Security – Wireless – Automation

MPC10E Deepdive

An HPE Juniper Networking Techpost

Deepaktr

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1 MPC10E Deepdive

Deepak Tripathi - 03/13/2023

A detailed view of the MPC10E line cards used in MX240, MX480 and MX480.

MX240/480/960 product family has been Juniper's flagship product for many years. It continues to delight our customers by protecting their investments through continuous capacity upgrades without having to rip and replace the entire system. MPC10E is the latest entry in the long list of Line Cards/Modular Physical Interface Cards (MPC) supported in this family.

1.1 Introduction



Figure 1: MPC10E Line Cards

MPC10E comes in two variants, MPC10E-10c and MPC10E-15c, which can deliver 1Tbps and 1.5Tbps of throughput capacity, respectively. Each MPC10E supports different port speeds, including 400G, making it a true multi-rate or MRATE line card. Here is the summary of the speeds supported:

Speed	MPC10E-10c	MPC10E-15c
10GE	40 (with 4x10GE Breakout)	60 (with 4x10GE Breakout)
25GE	40 (with 4x10GE Breakout)	60 (with 4x10GE Breakout)
40GE	10	15
100GE	10	15
400GE	2	3

Please find the complete list of supported optics from the Hardware Compatibility Tool:

MPC10E family of line cards runs on the fifth generation of Trio. Each line card will have two or three of these ASICs, depending on the MPC type.

1.2 Trio5 Architecture



Figure 2: Trio5 Picture

Trio5 is a 500Gbps chipset in 16nm design. Just like previous generations, Trio5 is optimized for edge deployments. The built-in crypto engine is introduced for the first time in the Trio family. This enables inline MACSec at all port speeds starting from 10GE to 100GE. 400GE MACSec is possible through Trio6. Please refer to my blog on LC9600 to get more details.

Trio5 continues to support a large scale of advanced edge features supported in the previous revisions of Trio. The same has been discussed in detail in the LC480 TechPost. It supports a high number of queues and virtually unlimited firewall filters along with prefix lists and tunnels. It supports HQoS, Inline services such as NAT, 6RD, MAP-E, and Jflow/NetFlow along with telemetry, flex-filters and up to 16 label depth to enable advanced use cases of traffic engineering with segment routing (SR/SRv6).

Trio5 is the packet forwarding engine or PFE in the MPC10E family of line cards. To optimize power and space many design considerations have been taken into account. First and foremost is the memory. Trio5 replaces

the Hybrid Memory Cube (HMC) and in-house high-performance memory with High Bandwidth Memory (HBM), which is used for packet processing and WAN queuing systems. Another important difference from the previous version is the inclusion of HBM memory and Trio ASIC via a silicon interposer in 2.5D packaging design. This design reduces the bus length enabling better power performance and minimizes board area on the line cards. It helps in packing more Trio5 ASICS on a single-line card, thus improving the per-slot throughput by more than three times compared to line cards based on previous versions.

The Crypto engine has been introduced for the first time in the Trio family. It enables building power-efficient systems/line cards that won't require an external component such as PHY to secure the ethernet connections between routers and switches.

The size of on-chip memory (OCMEM) in Trio5 has been increased, which is used for on-chip delay bandwidth buffers and the WAN queuing functions.

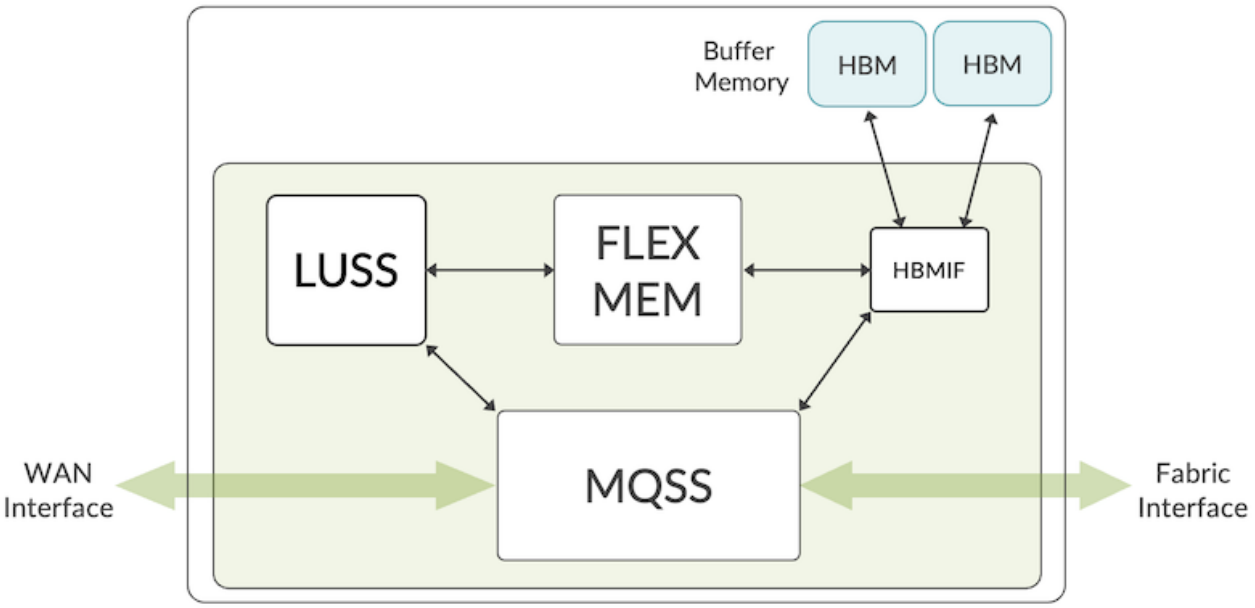


Figure 3: Trio5 Architecture

High Bandwidth Memory (HBM) is used as off-chip memory for delay bandwidth buffer and high scale flow table (Jflow) storage.

Each Trio5 has the following main components:

1. Lookup SubSystem (LUSS) provides all packet processing functions such as route/label lookup, firewall, and multi-field packet classification. This subsystem holds an array of Packet Processing Engines (PPE) to perform these functions. Trio5 and Trio4 each have 96 PPE, but they run at a higher clock frequency to handle the increase in throughput in Trio5.
2. Memory and Queuing SubSystem (MQSS) provides data paths and rich queuing functionality. It acts as an interface between WAN and Fabric. It has a pre-classifier where packets are classified as low/high priority. Unlike initial generations of Trio where an eXtended Queuing SubSystem (XQSS) was used to provide rich queuing functionality, Trio5 integrates this function within the MQSS block. This helps in the reduction of foot print and improves the power performance without any compromise on functionality.
3. HBMIF is the interface to HBM mem and on-chip FlexMem

4. FlexMem is the on-chip memory that is used for WAN queuing data structures and on-chip delay bandwidth storage.

1.2.1 Life of Packet inside Trio5

The life of the packet inside Trio5 is similar to the one described in Trio6 TechPost

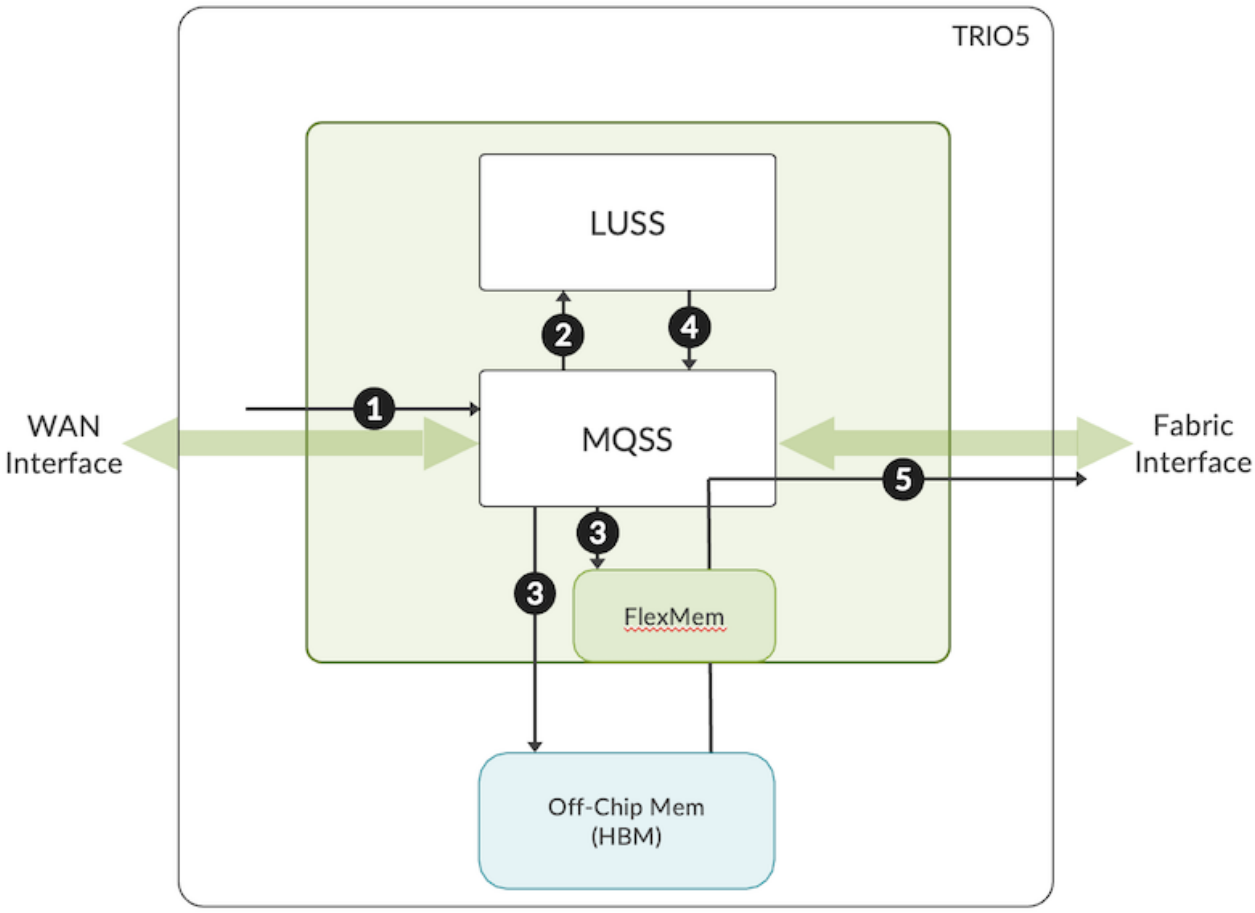


Figure 4: Trio5 Life of a Packet

1. A packet is received on the MQSS block either from the WAN interface or from the Fabric interface. Pre-classification decides priority. The main purpose of pre-classification is to make sure that high-priority control traffic is protected even if the PFE is oversubscribed.
2. If the incoming packet size is less of equal to 224 bytes, the complete packet will be sent to the LUSS. If the incoming packet is larger than 224 bytes, the packet is split into HEAD (192 bytes) and TAIL. A reorder context and a reorder ID is created. Then, HEAD is sent to LUSS for processing, such as the route/label lookup function.
3. The TAIL of the packet is either sent to on-chip SRAM (FlexMem) or off-chip HBM.
4. The incoming packet gets processed in the PPEs in LUSS. Once LUSS has finished processing the modified packet or HEAD is sent back to MQSS. Here reorder entry of the packet is validated and once it becomes eligible it is sent to the fabric scheduler based on the priority queues.
5. Once the packet becomes eligible to be sent out of PFE, the content will be read from FlexMem or HBM and the packet will be sent out via the WAN/Fabric interface.

1.3 MPC10E Architecture

The number of Packet Forwarding Engines (PFE) in MPC10E depends on the flavour. If it is a 1Tbps MPC10E-10c line card, then it will have two Trio5s, and if it is MPC10e-15c, then it will have three Trio5s. The 15c variant will have a slightly different approach in connecting to the fabric. More on this will be discussed in the next section. Leaving this aside, both the line cards have similar architecture.

Here is MPC10E-15c installed in the MX480 chassis.

```
regress@mx480-MPC10E> show chassis fpc 1 detail
Slot 1 information:
  State Online
  Temperature 43 degrees C / 109 degrees F
  Total CPU DRAM 32768 MB
  Total HBM 24576 MB
  FIPS Capable True
  FIPS Mode False
  Start time 2022-11-15 20:37:55 PST
  Uptime 19 days, 4 hours, 19 minutes, 24 seconds
  Max power consumption 785 Watts

PFE Information:

PFE Power ON/OFF Bandwidth SLC
  0 ON 500G
  1 ON 500G
  2 ON 500G
```

The faceplate of the line card will have 15/10G optical slots. Each of these slots can, by default, support 100G speed. Only the fifth, tenth and fifteenth ports can support 400G speed.

```
regress@mx480-MPC10E> show chassis pic pic-slot 0 fpc-slot 0
FPC slot 0, PIC slot 0 information:
  Type MRATE-5xQSFPP
  State Online
  PIC version 0.0
  Uptime 2 days, 15 hours, 27 minutes, 58 seconds
PIC port information:
                   Fiber Xcvr vendor Wave- Xcvr JNPR MSA
  Port Cable type type Xcvr vendor part number length Firmware Rev Version
  0 40GBASE SR4 MM AVAGO AFBR-79EQDZ-JU1 850 nm 0.0 REV 01 SFF-8436 ver n/a
  4 400GBASE-FR4 SM JUNIPER-1W2 740-085349 1301 nm 1.0 REV 01 CMIS 3.0

Port speed information:
  Port PFE Capable Port Speeds
  0 0 4x10GE, 4x25GE, 40GE, 100GE
  1 0 4x10GE, 4x25GE, 40GE, 100GE
  2 0 4x10GE, 4x25GE, 40GE, 100GE
  3 0 4x10GE, 4x25GE, 40GE, 100GE
  4 0 4x10GE, 4x25GE, 40GE, 100GE, 4x100GE, 400GE
```

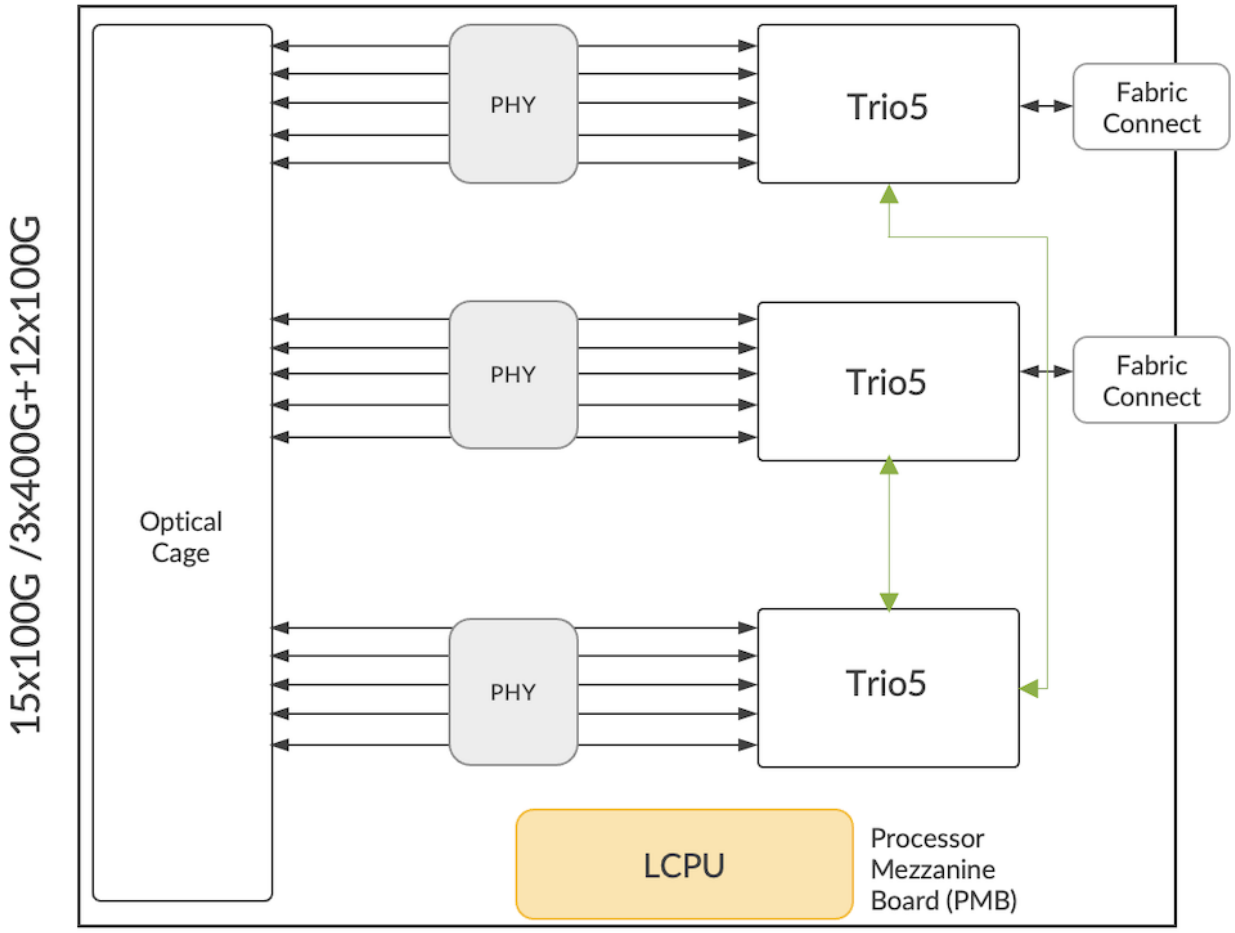


Figure 5: MPC10E-15C Architecture

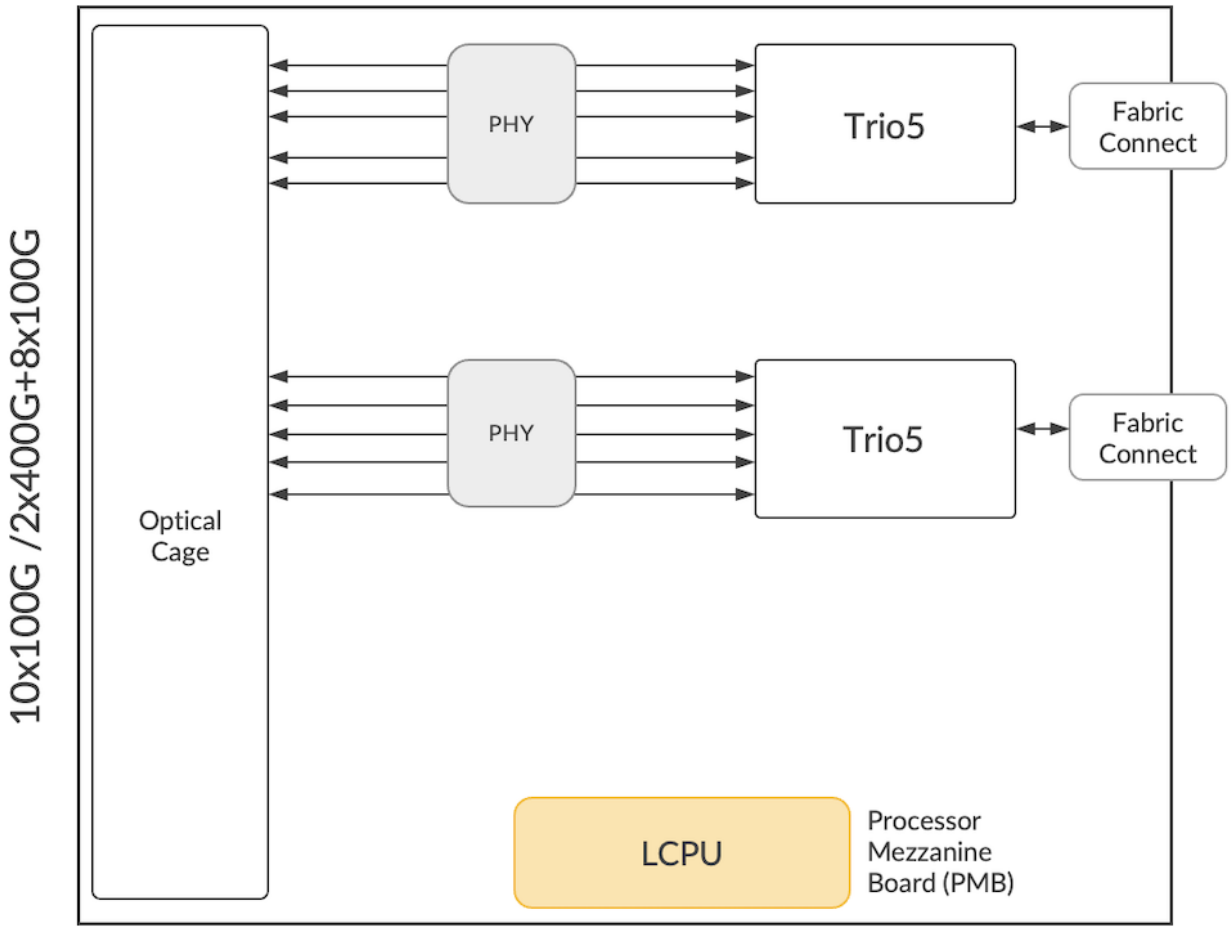


Figure 6: MPC10E-10C Architecture

Line Card CPU (LCPU) is installed on a Processor Mezzanine Board (PMB) and serves for the standard Line card and PFE management functions. Eight core LCPU supports the bandwidth requirements of both the control and data plane. The LCPU runs the control packets and maintains other functions, such as:

- Update local route tables
- LOG, SYSLOG
- SFLOW
- JFLOW
- MACSec key exchanges
- Other bandwidth-intensive applications such as protocol session traffic, exception traffic handling, ARP, IPv4/IPv6 options etc.

There are 28 usable SerDes lanes between the ASIC and the WAN. But only 20 will be active at any given time, providing an aggregate bandwidth of 500Gbps. These SerDes lanes run at different speeds to support WAN interfaces from 10G to 400G.

MPC10E uses port profiles to manage the PIC ports. Since oversubscription is not supported on MPC10E a port profile selects a set of ports active in a PIC and the port speed. Here are the config options.

```
regress@mx480-MPC10E# set chassis fpc 1 pic 0 pic-mode ?
Possible completions:
  100G 100GE mode
  10G 10GE mode
```

```
  1G 1GE mode
  25G 25GE mode
  400G 400GE mode
  40G 40GE mode
[edit]
set chassis fpc 1 pic 0 pic-mode 100G number-of-ports 5
regress@mx480-MPC10E# show chassis
fpc 1 {
   pic 0 {
      pic-mode 100G;
      number-of-ports 5;
   }
}
network-services enhanced-ip;
```

MPC10E also provides an option to control the speed at the port level. Any change in the port speed will not require a PIC bounce.

```
regress@mx480-MPC10E# set chassis fpc 1 pic 0 port 0 speed ?
Possible completions:
  100g Sets the interface mode to 100Gbps
  10g Sets the interface mode to 10Gbps
  1G 1GE-Gigabit Ethernet
  25g Sets the interface mode to 25Gbps
  400g Sets the interface mode to 400Gbps
  40g Sets the interface mode to 40Gbps
  oc12-stm4 OC12 or STM4
  oc3-stm1 OC3 or STM1
  oc48-stm16 OC48 or STM16
regress@mx480-01# show chassis
fpc 1 {
   pic 0 {
      port 0 {
         speed 100g;
      }
      port 4 {
         speed 400g;
      }
   }
}
network-services enhanced-ip;
```

The MPC10E-10c/15c will work with fabric card SCBE3 on MX240/480 and MX960 systems. Customers can continue to use most of the legacy line cards with this fabric, including 16x10GE line cards based on 1st gen Trio. Here is the list of hardware that will interoperate with MPC10E and SCBE3.

FRU	Interoperability
MPC1E/MPC2E/ICHIP DPC/MS-DPC	No
SCB/SCBE/SCBE2	No
RE-1300/RE-1200	No
16x10GE MPC	Yes, with new enhanced midplane
MPC3E/MPC4E/MPC5E	Yes
NG-MPC2E/NG-MPC3E	Yes
MPC7E-MRATE/MPC7E-10G	Yes
MS-MPC/MS-MIC	Yes
SCBE3	Yes
RE-S-1800	Yes
RE-S-X6	Yes

FRU	Interoperability
MX-SPC3	Yes

1.3.1 MPC10E-10c/15c and MX960 Fabric Interconnect

MX960 can have a maximum of three fabric cards. Each of these fabric cards will have one fabric chip. For the fabric interface of the MPC10E, only two Trio5 ASICs will connect to the backplane fabric, while the third "cascaded" Trio5 on the MPC10E-15c card will have its fabric interface connected to the other two Trio5 ASICs via their fabric cascade ports. This is design choice has been made for the MPC10E to interoperate with systems having older midplanes.

The two main ASICS have 32x PAM4 SerDes links, each connecting to the backplane fabric interface connector (24 active SerDes links per card in MX960). Each of those Trio5s will directly connect to a third Trio5 ASIC's fabric port via six lines of 52 Gbps PAM4 serdes from the cascaded port interface. This cascade port interface in the ASIC will accomplish a 2:1 weighted spray merge of traffic from the third ASIC, with half of the third Trio5 ASIC's WAN traffic going to each of the other two fabric-connected ASICs.

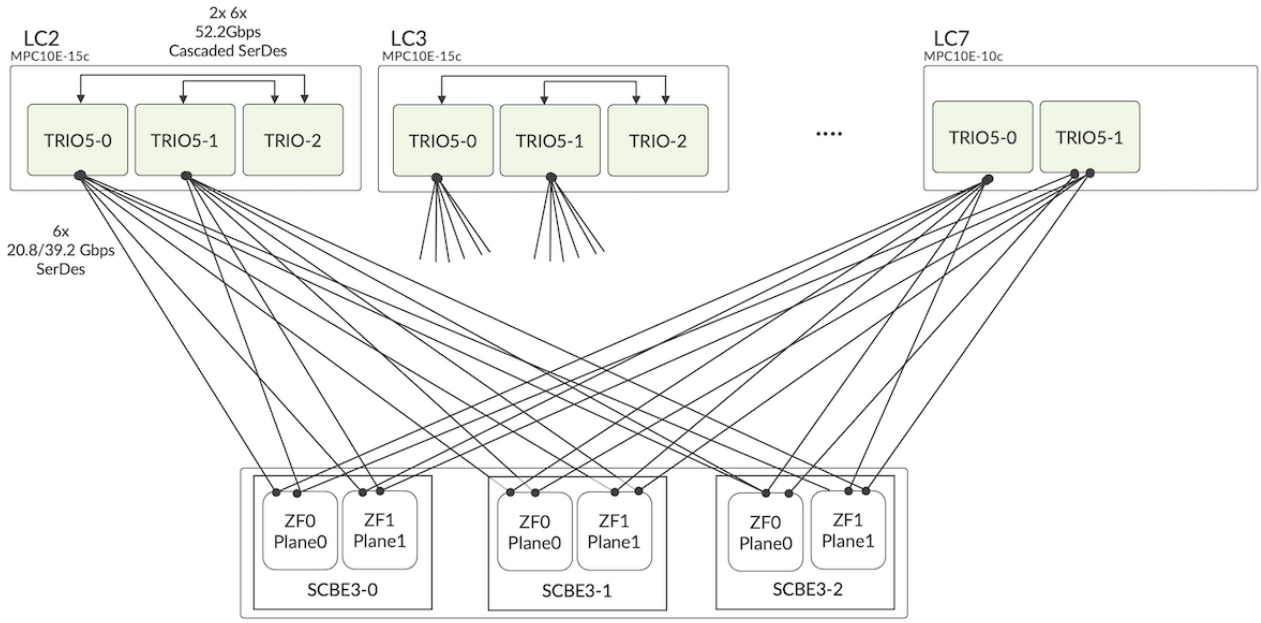


Figure 7: MPC10E and SCBE3 (MX960) Interconnect

MPC10E-10c has only two Trio5s, so the fabric cascade SerDes lanes will spray the traffic onto the fabric. This helps in achieving line-rate performance with two fabric cards in the case of MPC10E-10c.

As shown in the MPC10E and SCBE3 interconnect picture. There are a total of 6 planes available in the three fabric cards. All the planes are needed to be active to achieve line rate performance in a system with enhanced midplane for MPC10E-15c. For systems with legacy midplanes, the throughput performance will reduce. Here is table summarizing all these details.

MX960 Redundancy	Throughput – MPC10E-15c	Throughput – MPC10E-15c
3+0	1.5T	1T
2+1	1T	1T

There is one important consideration that has to be taken care of while installing an MPC10E line card. Due to cooling and power requirements, MX960 cannot have MPC10E in slots 0, 1 and 11. In a DC environment or at 25C, all the remaining slots of MX960 can have both variants of MPC10E. There are no such restrictions on MX480 and MX240.

Here is a table with the number of MPC10E supported under different environmental conditions.

Hardware SKU	Temperature	MX960	MX480	MX240
MPC10E-10C	25C	8 slotsSlots 0, 1, 11: N/A	6 slots	2 slots
	40C	8 slotsSlots 0, 1, 11: N/A	6 slots	2 slots
	55C	8 slotsSlots 0, 1, 11: N/A	6 slots	2 slots
MPC10E-15C	25C	8 slotsSlots 0, 1, 11: N/A	5 slots	2 slots
	40C	7 slotsSlots 0, 1, 11: N/A	4 slots	2 slots
	55C	Not Supported	Not Supported	Not Supported

```
regress@mx960-MPC10E> show chassis hardware
Hardware inventory:
Item Version Part number Serial number Description
Chassis JN124F4A4AFA MX960
Midplane REV 06 750-047849 ACRD6288 Enhanced MX960 Backplane
FPM Board REV 03 710-014974 ABDA6909 Front Panel Display
PDM Rev 03 740-013110 QCS18305095 Power Distribution Module
PEM 0 Rev 11 740-027760 QCS1828N068 PS 4.1kW; 200-240V AC in
PEM 1 Rev 11 740-027760 QCS1847N034 PS 4.1kW; 200-240V AC in
PEM 2 Rev 11 740-027760 QCS1828N00C PS 4.1kW; 200-240V AC in
PEM 3 Rev 11 740-027760 QCS1828N01K PS 4.1kW; 200-240V AC in
Routing Engine 0 REV 05 750-072925 CAPR8470 RE-S-2X00x6
Routing Engine 1 REV 05 750-072925 CARC2899 RE-S-2X00x6
CB 0 REV 32 750-070866 CAPT1368 Enhanced MX SCB 3
CB 1 REV 32 750-070866 CAPT2407 Enhanced MX SCB 3
FPC 3 REV 43 750-056519 CAJR8851 MPC7E 3D MRATE-12xQSFPP-XGE-XLGE-CGE
  CPU REV 20 750-057177 CAJP3829 SMPC PMB
  PIC 0 BUILTIN BUILTIN MRATE-6xQSFPP-XGE-XLGE-CGE
    Xcvr 2 REV 01 740-061409 1G3TQAA6060HR QSFP-100GBASE-LR4-T2
    Xcvr 5 REV 01 740-061409 1GTQA533098 QSFP-100GBASE-LR4-T2
  PIC 1 BUILTIN BUILTIN MRATE-6xQSFPP-XGE-XLGE-CGE
FPC 4 REV 30 750-028467 ZN1207 MPC 3D 16x 10GE
  CPU REV 10 711-029089 ZM7387 AMPC PMB
  PIC 0 BUILTIN BUILTIN 4x 10GE(LAN) SFP+
    Xcvr 0 REV 01 740-021308 AA1025A4UWU SFP+-10G-SR
  PIC 1 BUILTIN BUILTIN 4x 10GE(LAN) SFP+
    Xcvr 0 REV 01 740-031981 45T012402990 SFP+-10G-LR
  PIC 2 BUILTIN BUILTIN 4x 10GE(LAN) SFP+
    Xcvr 0 REV 01 740-011613 PAJ4J29 SFP-SX
  PIC 3 BUILTIN BUILTIN 4x 10GE(LAN) SFP+
FPC 5 REV 53 750-070395 CAPD0787 MPC10E 3D MRATE-15xQSFPP
  CPU REV 20 750-072571 CANF8715 FMPC PMB
  PIC 0 BUILTIN BUILTIN MRATE-5xQSFPP
    Xcvr 2 REV 01 740-061409 1G3TQAA61508X QSFP-100GBASE-LR4-T2
    Xcvr 3 M NON-JNPR FNS21161BBG QSFP-100GBASE-LR4
  PIC 1 BUILTIN BUILTIN MRATE-5xQSFPP
    Xcvr 2 REV 01 740-073093 1AMPA522037 QSFP+-40G-LR4
  PIC 2 BUILTIN BUILTIN MRATE-5xQSFPP
    Xcvr 0 REV 01 740-061409 1GTQA5330AT QSFP-100GBASE-LR4-T2
```

```
    Xcvr 1 REV 01 740-061409 1GTQA5330ED QSFP-100GBASE-LR4-T2
FPC 6 REV 36 750-053323 CAFT8776 MPC7E 3D 40XGE
  CPU REV 16 750-057177 CAFP4670 SMPC PMB
  PIC 0 BUILTIN BUILTIN 20x10GE SFPP
    Xcvr 0 REV 01 740-021308 943151A00527 SFP+-10G-SR
    Xcvr 1 REV 01 740-030658 AA1206AHWUG SFP+-10G-USR
    Xcvr 2 REV 01 740-021308 AA1043A8951 SFP+-10G-SR
    Xcvr 3 REV 01 740-021308 983152A00096 SFP+-10G-SR
  PIC 1 BUILTIN BUILTIN 20x10GE SFPP
FPC 8 REV 18 750-045372 CADW7020 MPCE Type 3 3D
  CPU REV 10 711-035209 CADV8102 HMPC PMB 2G
  MIC 0 REV 26 750-028392 CAAS5045 3D 20x 1GE(LAN) SFP
    PIC 0 BUILTIN BUILTIN 10x 1GE(LAN) SFP
      Xcvr 0 REV 01 740-011613 AGS0736G3GF SFP-SX
      Xcvr 9 REV 01 740-031469 17T446600290 SFP-LX10
    PIC 1 BUILTIN BUILTIN 10x 1GE(LAN) SFP
      Xcvr 3 REV 01 740-038291 PQ351CY SFP-T
      Xcvr 5 REV 01 740-031851 AM1226SXYXB SFP-SX
      Xcvr 6 REV 01 740-031851 AM1228SY6NR SFP-SX
Fan Tray 0 REV 08 740-031521 ACDB9008 Enhanced Fan Tray
Fan Tray 1 REV 08 740-031521 ACDB9042 Enhanced Fan Tray

{master}
regress@mx960-MPC10E> show chassis fpc pic-status
Slot 3 Online MPC7E 3D MRATE-12xQSFPP-XGE-XLGE-CGE
  PIC 0 Online MRATE-6xQSFPP-XGE-XLGE-CGE
  PIC 1 Online MRATE-6xQSFPP-XGE-XLGE-CGE
Slot 4 Online MPC 3D 16x 10GE
  PIC 0 Online 4x 10GE(LAN) SFP+
  PIC 1 Online 4x 10GE(LAN) SFP+
  PIC 2 Online 4x 10GE(LAN) SFP+
  PIC 3 Online 4x 10GE(LAN) SFP+
Slot 5 Online MPC10E 3D MRATE-15xQSFPP
  PIC 0 Online MRATE-5xQSFPP
  PIC 1 Online MRATE-5xQSFPP
  PIC 2 Online MRATE-5xQSFPP
Slot 6 Online MPC7E 3D 40XGE
  PIC 0 Online 20x10GE SFPP
  PIC 1 Online 20x10GE SFPP
Slot 8 Online MPCE Type 3 3D
  PIC 0 Online 10x 1GE(LAN) SFP
  PIC 1 Online 10x 1GE(LAN) SFP

{master}
regress@mX960-MPC10E>
```

The above output shows different generations of MPCs successfully operating in an MX960 chassis along with the required common components.

1.4 References

- MX960/480/240 Hardware Guide

https://www.juniper.net/documentation/us/en/hardware/mx960/mx960.pdf https://www.juniper.net/documentation/us/en/hardware/mx480/mx480.pdf https://www.juniper.net/documentation/us/en/hardware/mx240/mx240.pdf

1.5 Glossary

- FIB: Forwarding Information Base
- HBM: High Bandwidth Memory
- LCPU: Line Card CPU
- LUSS: Lookup Sub-System
- MCIF: Memory Control Interface
- MPC: Modular Port Concentrator

- MQSS: Memory and Queuing Sub-System:
- NRZ: Non-Return to Zero
- OCPMem: On-Chip Memory
- PCB: Printed Circuit Board
- PFE: Packet Forwarding Engine
- PIC: Physical Interface Cards
- PMB: Processor Mezzanine Board
- PPE: Packet Processing Engines
- QSFP-DD: Quad Small Form Factor Pluggable Double Density
- SerDes: Serializer/Deserializer
- SRAM: Static Random Access Memory
- SCBE3: Switch Control Board Card (gen3)
- XQSS: Extended Queuing Sub-System
- ZF: Chipset used in SFB Switch Fabric Cards

1.6 Acknowledgement

I want to express my gratitude to my mentor Nicolas Fevrier, Sr. Director, PLM, for the detailed reviews of the blog. I would also like to thank Eswaran Srinivasan, Distinguished Engineer and Vasily Mukhin for providing their valuable input.



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